

CONTENTS IN DETAIL

Introduction	xv
Project 1, Bagels: Deduce a secret three-digit number based on clues. <i>Practice using constants.</i>	1
Project 2, Birthday Paradox: Determine the probability that two people share the same birthday in groups of different sizes. <i>Use Python's datetime module.</i>	6
Project 3, Bitmap Message: Display a message on the screen configured by a 2D bitmap image. <i>Work with multiline strings.</i>	11
Project 4, Blackjack: A classic card game played against an AI dealer. <i>Learn about Unicode characters and code points.</i>	15
Project 5, Bouncing DVD Logo: Simulates the colorful bouncing DVD logo of decades past. <i>Work with coordinates and colorful text.</i>	23
Project 6, Caesar Cipher: A simple encryption scheme used thousands of years ago. <i>Convert between letters and numbers to perform math on text.</i>	29
Project 7, Caesar Hacker: A program to decrypt Caesar cipher messages without the encryption key. <i>Implement a brute-force cryptanalysis algorithm.</i>	33
Project 8, Calendar Maker: Create calendar pages for a given year and month. <i>Use Python's datetime module and the timedelta data type.</i>	36
Project 9, Carrot in a Box: A silly bluffing game between two players. <i>Create ASCII art.</i>	41
Project 10, Cho-Han: A gambling dice game from feudal Japan. <i>Practice using random numbers and dictionary data structures.</i>	47